Chapter 8

Inequalities and Limit Theorems

8.1 Two important inequalities

Sometimes a random variable, X, can have a distribution which is difficult to determine explicitly, but it is still possible to determine moments of X (e.g. by using linearity of expected value). In such situations, we can still get some information about *probabilities* associated with X by using probability inequalities based on moments of X.

8.1.1 Markov's inequality

Suppose X is a *non-negative* random variable and suppose $\varepsilon > 0$ is a constant. Then

$$\mathbb{P}(X \geqslant \varepsilon) \leqslant \frac{\mathbb{E}(X)}{\varepsilon}$$

Proof
We start by defining the indicator variable

$$I_\varepsilon = \begin{cases} 1 & \text{if } X \geqslant \varepsilon \\ 0 & \text{otherwise} \end{cases}$$

and note that $\varepsilon I_\varepsilon \leqslant X$, since $I_\varepsilon = 1$ only when $X \geq \varepsilon$.
So it follows from the properties of expected value that $\mathbb{E}(\varepsilon I_\varepsilon) \leqslant \mathbb{E}(X)$. $(*)$
Next, we compute:

$$\mathbb{E}(\varepsilon I_\varepsilon) = \varepsilon \mathbb{E}(I_\varepsilon) = \varepsilon \mathbb{P}(I_\varepsilon = 1) = \varepsilon \mathbb{P}(X \geqslant \varepsilon)$$

So, substituting $\varepsilon \mathbb{P}(X \geqslant \varepsilon)$ for $\mathbb{E}(\varepsilon I_\varepsilon)$ in $(*)$, we obtain

$$\varepsilon \mathbb{P}(X \geqslant \varepsilon) \leqslant \mathbb{E}(X)$$

and the result follows.

Remark It is important to note that Markov's inequality can only be used when we know that the random variable X takes *non-negative* values (can you see where we use the fact that X is non-negative in the proof above?). So be careful when you use Markov's inequality and always check that X is a non-negative random variable first.

Example 8.1. Suppose $X_1, X_2, \ldots, X_5$ are independent, identically distributed random variables, each being exponentially distributed with mean 2. Use Markov's inequality to bound the probability that the sum of these random variables is greater than 13.

Solution: Let $S = X_1 + X_2 + \cdots + X_5$. Then S is a non-negative random variable, with $E[S] = \sum_{i=1}^{5} E[X_i] = 10$. Applying Markov's inequality,

$$\begin{aligned} \mathbb{P}(S > 13) &= \mathbb{P}(S \geqslant 13) && \text{(since } S \text{ is a continuous r.v.)} \\ &\leqslant \frac{E[S]}{13} && \text{(Markov's inequality)} \\ &= \frac{10}{13} \end{aligned}$$

Note that we did not need to know the distribution of the random variable S to derive the above bound.

Although Markov's inequality can only be used when X is a non-negative random variable, we can apply it to obtain an inequality that can be used for more general random variables.

8.1.2 Chebyshev's inequality

Suppose X is a random variable with $\mathbb{E}(X) = \mu$ and $\mathbb{V}\text{ar}(X) < \infty$, and suppose $\varepsilon > 0$ is a constant. Then

$$\mathbb{P}(|X - \mu| \geqslant \varepsilon) \leqslant \frac{\mathbb{V}\text{ar}(X)}{\varepsilon^2}$$

Proof. The inequality follows from an application of Markov's inequality to the *non-negative* random variable $(X - \mu)^2$ as follows:

$$\mathbb{P}(|X - \mu| \geqslant \varepsilon) = \mathbb{P}((X - \mu)^2 \geqslant \varepsilon^2) \leqslant \frac{\mathbb{E}\big((X - \mu)^2\big)}{\varepsilon^2} = \frac{\mathbb{V}\text{ar}(X)}{\varepsilon^2}$$

remembering that $\mathbb{E}((X - \mu)^2) = \mathbb{V}\text{ar}(X)$, by definition of $\mathbb{V}\text{ar}(X)$.

Remark. In statistical applications we are often interested in determining the probability that a random variable X is 'far away' (i.e. more than ε) from its mean μ. Chebyshev's inequality gives us a way to bound this probability.

Example 8.2. Suppose that $X \sim \Gamma(200, 2)$. Use Chebyshev's inequality to find a lower bound on the probability $\mathbb{P}(90 < X < 110)$.

Solution: For $X \sim \Gamma(200, 2)$, we have $E[X] = \frac{200}{2} = 100$, and $\mathbb{V}\text{ar}(X) = \frac{200}{2^2} = 50 < \infty$. Applying Chebyshev's inequality with $\varepsilon = 10$, we obtain

$$\mathbb{P}(|X - 100| \geqslant 10) \leqslant \frac{50}{10^2} = \frac{1}{2}$$

and so

$$\mathbb{P}(90 < X < 110) = \mathbb{P}(|X - 100| < 10) = 1 - \mathbb{P}(|X - 100| \geqslant 10) \geqslant \frac{1}{2}$$

There is probability at least 0.5 that the random variable X is within 10 of its mean, by Chebyshev's inequality.

Application of Chebyshev's Inequality: Weak Law of Large Numbers
Suppose $X_1, X_2, \ldots$ are independent, identically distributed random variables with $\mathbb{E}(X) = \mu$ and $\mathbb{V}\text{ar}(X) = \sigma^2 < \infty$ and let $S_n = X_1 + \cdots + X_n$. Then for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\Big(\Big| \underbrace{\frac{S_n}{n}}_{\text{sample mean}} - \mu\Big| \geqslant \varepsilon\Big) = 0.$$

Proof
First, we note that

$$\begin{aligned}
\mathbb{E}\Big(\frac{S_n}{n}\Big) &= \frac{\mathbb{E}(X_1 + \cdots + X_n)}{n} \\
&= \frac{\mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n)}{n} && \text{(property of expectation)} \\
&= \frac{\mu + \cdots + \mu}{n} = \mu && \text{(since the variables have the same mean)}
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{V}\text{ar}\Big(\frac{S_n}{n}\Big) &= \frac{1}{n^2}\mathbb{V}\text{ar}(X_1 + \cdots + X_n) && \text{(property of variance)} \\
&= \frac{1}{n^2}\big(\mathbb{V}\text{ar}(X_1) + \cdots + \mathbb{V}\text{ar}(X_n)\big) && \text{(since } X_1, \ldots, X_n \text{ are independent)} \\
&= \frac{1}{n^2}(\sigma^2 + \cdots + \sigma^2) = \frac{\sigma^2}{n} && \text{(since the variables have the same variance)}
\end{aligned}$$

So, by Chebyshev:

$$0 \leqslant \mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| \geqslant \varepsilon\right) \leqslant \frac{\mathbb{V}\text{ar}(\frac{S_n}{n})}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \qquad (*)$$

It follows from $(*)$ that $\lim_{n\to\infty} \mathbb{P}\Big(\big|\frac{S_n}{n} - \mu\big| \geqslant \varepsilon\Big) = 0$.
Remark The Weak Law of Large Numbers is used in various statistical applications.

8.2 Central Limit Theorem

One of the most important results in probability theory and statistics is the Central Limit Theorem.

Central Limit Theorem (CLT)
Suppose that $X_1, X_2, \ldots$ are independent and identically distributed random variables with $\mathbb{E}(X_i) = \mu$, $\mathbb{V}\text{ar}(X_i) = \sigma^2 < \infty$, and let $S_n = X_1 + \ldots X_n$ for $n = 1, 2, \ldots$
Then for *any* $a < b$,

$$\lim_{n\to\infty} \mathbb{P}\left(a \leqslant \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leqslant b\right) = \mathbb{P}\left(a \leqslant Z \leqslant b\right) \quad (*) \qquad \text{where } Z \sim \mathrm{N}\left(0, 1\right)$$

Proof: Beyond the scope of this course (requires use of *generating functions*).

Remark. Another way to describe the Central Limit Theorem is to say that

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \quad \textit{converges in distribution to} \quad Z \sim \mathrm{N}\left(0, 1\right)$$

as $n \to \infty$. We say this because equality $(*)$ above tells us that probabilities for $\frac{S_n - n\mu}{\sqrt{n\sigma^2}}$ converge to probabilities for $Z \sim \mathrm{N}\left(0, 1\right)$.

Remark. Typically we use the Central Limit Theorem to justify the statement:
For n *large* and for independent and identically distributed random variables $X_1, X_2, \ldots, X_n$ with $\mathbb{E}(X_i) = \mu$, $\mathbb{V}\mathrm{ar}(X_i) = \sigma^2 < \infty$, and $S_n = X_1 + \ldots X_n$, we have, by the Central Limit Theorem,

$$\mathbb{P}\left(a \leqslant \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leqslant b\right) \approx \mathbb{P}(a \leqslant Z \leqslant b)$$

where $Z \sim \mathrm{N}(0,1)$.

Central Limit Theorem applied to Binomial distribution
Suppose that $0 < p < 1$ and that X_n is a Binomial $\mathrm{Bin}(n,p)$ random variables for $n = 1, 2, \ldots$. Then for any $a < b$ we have

$$\lim_{n\to\infty} \mathbb{P}\left(a \leqslant \frac{X_n - np}{\sqrt{np(1-p)}} \leqslant b\right) = \mathbb{P}(a \leqslant Z \leqslant b) \qquad \text{where } Z \sim \mathrm{N}(0,1)\,.$$

Proof. Recall that the Binomial variable X_n counts the number of successes in n independent trials with $\mathbb{P}(\text{success}) = p$. So we can write $X_n = I_1 + I_2 + \cdots + I_n$ where

$$I_k = \begin{cases} 1 & k^{\text{th}} \text{ trial is success} \\ 0 & \text{otherwise} \end{cases}$$

and the indicator variables $I_1, I_2, \ldots$ are independent. We also know that $\mathbb{E}(I_k) = p$ and $\mathbb{V}\mathrm{ar}(I_k) = p(1-p)$. So, by the Central Limit Theorem,

$$\lim_{n\to\infty} \mathbb{P}\left(a \leqslant \frac{X_n - np}{\sqrt{np(1-p)}} \leqslant b\right) = \lim_{n\to\infty} \mathbb{P}\left(a \leq \frac{I_1 + \cdots + I_n - np}{\sqrt{np(1-p)}} \leqslant b\right) = \mathbb{P}(a \leqslant Z \leqslant b)$$

Remark. Typically we use the above result to justify the statement:
For $X_n \sim \mathrm{Bin}(n,p)$ with n *large*, we have, by the Central Limit Theorem,

$$\mathbb{P}\left(a \leqslant \frac{X_n - np}{\sqrt{np(1-p)}} \leqslant b\right) \approx \mathbb{P}(a \leqslant Z \leqslant b)$$

where $Z \sim \mathrm{N}(0,1)$.
This is referred to as the **Normal Approximation to the Binomial**.

Example 8.3. 100 new light bulbs have independent lifetimes that are exponentially distributed with mean 5 weeks. The light bulbs are used one at a time and when a bulb fails, it is immediately replaced. Suppose the first bulb is installed at time $t = 0$. What is the (approximate) probability that all of the 100 new bulbs will have been used and have failed before the end of 550 weeks?

Solution: To answer this question, we begin by defining some notation. For $i = 1, 2, ..., 100$, let X_i equal the lifetime of ith light bulb. From the information given in the problem, we know that the variables $X_1, X_2, \ldots, X_{100}$ are *independent*. For $k = 1, 2, \ldots, 100$, we know that X_i is exponentially distributed with mean 5 weeks, so $X_i \sim \mathrm{Exp}(1/5)$, and $\mathbb{V}\mathrm{ar}(X_i) = 25$.

The time until all of the light bulbs have been used and have failed is $X_1 + X_2 + \cdots + X_{100}$ and we want to compute

$$\mathbb{P}(X_1 + X_2 + \cdots + X_{100} \leqslant 550)$$

Since $X_1, X_2, \ldots, X_n$ are i.i.d., we can use the Central Limit Theorem (CLT) to obtain

$$\begin{aligned}
\mathbb{P}(X_1 + X_2 + \cdots + X_{100} \leqslant 550) &= \mathbb{P}\left(\frac{X_1 + X_2 + \cdots + X_{100} - 100\mathbb{E}(X_i)}{\sqrt{100\mathbb{V}\mathrm{ar}(X_i)}} \leqslant \frac{550 - 100\mathbb{E}(X_i)}{\sqrt{100\mathbb{V}\mathrm{ar}(X_i)}}\right) \\
&\approx \mathbb{P}\left(Z \leqslant \frac{50}{\sqrt{2500}}\right) \qquad \text{by the CLT, with } Z \sim \mathrm{N}(0,1) \\
&= \mathbb{P}(Z \leqslant 1) = 0.8413 \qquad \text{from the Statistical Tables}
\end{aligned}$$

8.3 Continuity correction; more examples of CLT

The Central Limit Theorem allows us to use Normal distribution probabilities to approximate both continuous distributions and discrete distributions. When approximating a *discrete* distribution using the (*continuous*) Normal distribution, there is an important refinement, called a *continuity correction*.

Example 8.4. Suppose that $X \sim B(200, 0.5)$, and we want to approximate $\mathbb{P}(X = 105)$ using the CLT.

Note that **because X takes only integer values,** $\mathbb{P}(X = 105) = \mathbb{P}(104.5 \leq X \leq 105.5)$.

Now we have $E(X) = 100$, $\mathbb{V}\mathrm{ar}(X) = 50$, and so by the CLT,

$$\begin{aligned}
\mathbb{P}(X = 105) &= \mathbb{P}(104.5 \leqslant X \leqslant 105.5) \qquad \text{(continuity correction)} \\
&= \mathbb{P}\left(\frac{104.5 - 100}{\sqrt{50}} \leqslant \frac{X - 100}{\sqrt{50}} \leqslant \frac{105.5 - 100}{\sqrt{50}}\right) \\
&\approx \mathbb{P}(0.64 \leqslant Z \leqslant 0.78) \qquad \text{where } Z \sim N(0,1), \text{ by CLT} \\
&= \Phi(0.78) - \Phi(0.64) \\
&= 0.7823 - 0.7389 \qquad \text{(from the Statistical Tables)} \\
&= 0.0434
\end{aligned}$$

More generally, if X is an integer-valued random variable, then for any integer values a, b, we have

$$\mathbb{P}(a \leqslant X \leqslant b) = \mathbb{P}(a - 0.5 \leqslant X \leqslant b + 0.5)$$

Note that the interval $\{a \leqslant X \leqslant b\}$ *includes* both end-points, so we *extend* the interval at both ends. Compare with

$$\mathbb{P}(a < X < b) = \mathbb{P}(a + 0.5 \leqslant X \leqslant b - 0.5)$$

In this case, the interval $\{a < X < b\}$ *excludes* both end-points, so we *contract* the interval at both ends.

Example 8.5. An electronic instrument functions if at least 40 of its 70 components are functioning. Components function independently and the lifetime of each component is exponentially distributed with mean 100 months (so $\lambda = 1/100$). What is the (approximate) probability that the instrument is still functioning after 80 months?

Define Y to be the number of components that are still functioning after 80 months. Since the components operate independently and all have the same lifetime distribution, the random variable Y is Binomial, $Y \sim \mathrm{Bin}(70, p)$, where $p = \mathbb{P}(\text{a component is still functioning after 80 months})$. Since the lifetime, X, of a component is exponentially distributed with $\lambda = 1/100$, we have

$$p = \mathbb{P}(X > 80) = e^{-80/100} = e^{-0.8}$$

Then we have

$\mathbb{P}$(instrument functioning after 80 months) $=\mathbb{P}(Y \geqslant 40) = \mathbb{P}(Y \geqslant 39.5)$

since Y can take only integer values, and the interval $\{Y \geqslant 40\}$ includes the value 40, so we have used a *continuity correction* by extending the interval to $\{Y \geqslant 39.5\}$.
Now

$$\mathbb{P}(Y \geqslant 39.5) = \mathbb{P}\left(\frac{Y - 70p}{\sqrt{70p(1-p)}} \geqslant \frac{39.5 - 70p}{\sqrt{70p(1-p)}}\right)$$

$$\approx \mathbb{P}\left(Z \geqslant \frac{39.5 - 70e^{-0.8}}{\sqrt{70e^{-0.8}(1-e^{-0.8})}}\right) \qquad \text{(CLT for Binomial, with } Z \sim \mathrm{N}(0,1)\text{)}$$

$$= \mathbb{P}(Z \geqslant 1.93) = 1 - \Phi(1.93) = 0.0268 \qquad \text{(from the Statistical Tables)}$$

Example 8.6. Suppose that we wish to estimate the proportion, p, of voters who intend to vote for Party A in an upcoming election. To estimate p, we select n people at random (without replacement) from the population of registered voters. Let

$$X_i = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ person selected intends to vote for Party A} \\ 0 & \text{otherwise} \end{cases}$$

We note that $\mathbb{E}(X_i) = p$ and $\mathbb{V}\mathrm{ar}(X_i) = p(1-p)$ and since the population is *very large*, we will *assume* that the variables $X_1, X_2, \ldots$ are independent.

To estimate the population proportion p, we will use the sample mean $\bar{X}(n) = \frac{X_1 + \cdots + X_n}{n}$, where n is the size of the selected sample.

Question: How large should n be to ensure that

$$\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big) \geqslant 0.95?$$

Answer: We begin by using the Central Limit Theorem to estimate $\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big)$:

$$\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big)$$

$$= \mathbb{P}\big(-0.05 \leqslant \bar{X}(n) - p \leqslant 0.05\big) \qquad \text{(algebra)}$$

$$= \mathbb{P}\big(-0.05n \leqslant X_1 + \cdots + X_n - np \leqslant 0.05n\big) \qquad \text{(since } n\bar{X}(n) = X_1 + \cdots + X_n\text{)}$$

$$= \mathbb{P}\left(\frac{-0.05n}{\sqrt{np(1-p)}} \leqslant \frac{X_1 + \cdots + X_n - np}{\sqrt{np(1-p)}} \leqslant \frac{0.05n}{\sqrt{np(1-p)}}\right)$$

$$\geqslant \mathbb{P}\left(\frac{-0.05\sqrt{n}}{\sqrt{0.25}} \leqslant \frac{X_1 + \cdots + X_n - np}{\sqrt{np(1-p)}} \leqslant \frac{0.05\sqrt{n}}{\sqrt{0.25}}\right) \qquad \text{(since } p(1-p) \leqslant 0.25\text{)}$$

$$\approx \mathbb{P}(-0.1\sqrt{n} \leqslant Z \leqslant 0.1\sqrt{n}) \qquad \text{(by the Central Limit Theorem, with } Z \sim N(0,1)\text{)}$$

Now, we want $\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big) \geqslant 0.95$ and, from the calculation above, we can see that this will be achieved if $\mathbb{P}(-0.1\sqrt{n} \leqslant Z \leqslant 0.1\sqrt{n}) = 0.95$. Using Table 5 from the Cambridge Statistical Tables, we see that for the standard normal variable Z,

$$\mathbb{P}(-1.96 \leqslant Z \leqslant 1.96) = 0.95.$$

So, $\mathbb{P}(-0.1\sqrt{n} \leqslant Z \leqslant 0.1\sqrt{n}) = 0.95$ if and only if $0.1\sqrt{n} = 1.96$. Solving $0.1\sqrt{n} = 1.96$, we obtain $n = 385$, and so a sample size $n = 400$ would be large enough to achieve $\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big) \geqslant 0.95$.

8.4 Poisson approximation to the binomial distribution

A different distributional approximation (not based on the Central Limit Theorem) is the following.
Suppose $\lambda > 0$ and for $n = 1, 2, \ldots$ let $Y_n \sim \text{Bin}(n, \frac{\lambda}{n})$.
Then the variables $Y_1, Y_2, \ldots$ converge in *distribution* to a Poisson variable $X \sim \text{Poi}(\lambda)$.

Remark: Convergence of $Y_1, Y_2, ...$ to X in distribution means that

$$\lim_{n\to\infty} \mathbb{P}(Y_n = k) = \mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Proof. Since $Y_n \sim B\left(n, \frac{\lambda}{n}\right)$, then for $k \leq n$ we have

$$\begin{aligned}
\mathbb{P}(Y_n = k) &= \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k} \\
&= \frac{n!}{(n-k)!\,k!}\left(\frac{\lambda^k}{n^k}\right)\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^k} \\
&= \frac{\lambda^k}{k!}\left(\frac{n(n-1)(n-2)\cdots(n-k+1)}{n^k}\right)\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^k} \\
&= \frac{\lambda^k}{k!}\,1\cdot\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right)\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^k} \qquad (*)
\end{aligned}$$

Now as $n \to \infty$,

$$\begin{aligned}
1\cdot\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) &\to 1 \\
\left(1-\frac{\lambda}{n}\right)^k &\to 1 \\
\left(1-\frac{\lambda}{n}\right)^n &\to e^{-\lambda}
\end{aligned}$$

Substituting into $(*)$, we obtain

$$\lim_{n\to\infty} \mathbb{P}(Y_n = k) = \frac{\lambda^k}{k!}e^{-\lambda}$$

as required.

Example 8.7. National Lottery. To win the UK National Lottery, the 6 numbers on your ticket have to match the 6 balls drawn from 59 balls, numbered $1, 2, \ldots, 59$. There are $\binom{59}{6}$

different possible subsets of size 6 from the 59 balls and all of these possible subsets are assumed to be equally likely to be selected, so

$$\mathbb{P}(\text{win}) = \frac{1}{\binom{59}{6}} = \frac{1}{45\,057\,474} \approx \frac{1}{N}$$

where $N = 45$ million.

Now suppose that $2N$ lottery tickets are sold and let Y equal the number of jackpot winners. What is the probability that there are 3 jackpot winners?

Assuming the numbers on the $2N$ lottery tickets are independent, we have

$$Y \sim \text{Bin}(2N, 1/N) = \text{Bin}(2N, 2/2N)$$

By the Poisson approximation to the Binomial, we have that probabilities for the Binomial variable Y can be approximated by probabilities for X, where X is a Poisson random variable with $\lambda = (2N) \cdot \left(\frac{2}{2N}\right) = 2$. So

$$\mathbb{P}(Y = 3) \approx \mathbb{P}(X = 3) = \frac{e^{-2} \cdot 2^3}{3!} = 0.18$$

Note: here we are approximating a *discrete* distribution (Binomial) by another *discrete* distribution (Poisson), so a continuity correction would **not** be appropriate.